CSCI 49376 Assignment 1

Relational Database Design and Normalization of a Large Dataset · Big Data Technology · Prof. Alex Liew

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Dataset	Option 2: Diabetes 130-US Hospitals for Years 1999–2008
This section	Parts 1–4: dataset analysis, normalization, ER diagram, relational schema

Part 1. The Original Dataset

Item	Answer
1. Dataset selected	Diabetes 130-US Hospitals for Years 1999–2008 (Kaggle: brandao/diabetes; Strack et al., *BioMed Research International*, 2014)
2. CSV file used	`diabetic_data.csv`
3. Total rows	**101,766** (one row per inpatient encounter)
4. Total columns	**50**. As instructed, 24 are ignored (23 medication columns and `change`), leaving **26** attributes in the design.

The data covers ten years of diabetic inpatient encounters at 130 U.S. hospitals. Each row is one hospital stay: who the patient was, how they were admitted and discharged, what was done during the stay, up to three diagnoses, two lab results, and whether the patient was readmitted.

5–6. All attributes in the original CSV

#	Attribute	Description	Design
1	`encounter_id`	Unique identifier of one hospital stay (encounter)	Used
2	`patient_nbr`	Unique identifier of a patient; one patient can have several encounters	Used
3	`race`	Patient race: Caucasian, AfricanAmerican, Hispanic, Asian, Other	Used
4	`gender`	Female, Male, Unknown/Invalid	Used
5	`age`	Age group at the time of the encounter, 10-year bands: [0-10) … [90-100)	Used
6	`weight`	Weight band in pounds, e.g. [75-100)	Used
7	`admission_type_id`	Code for the type of admission (e.g. emergency, urgent, elective)	Used
8	`discharge_disposition_id`	Code for where the patient went at discharge (e.g. home, transferred, expired)	Used
9	`admission_source_id`	Code for where the patient was admitted from (e.g. physician referral, emergency room)	Used
10	`time_in_hospital`	Days between admission and discharge (1–14)	Used
11	`payer_code`	Insurance/payer code (e.g. MC = Medicare)	Used
12	`medical_specialty`	Specialty of the admitting physician	Used
13	`num_lab_procedures`	Number of lab tests performed during the encounter	Used
14	`num_procedures`	Number of procedures other than lab tests	Used
15	`num_medications`	Number of distinct medications administered	Used
16	`number_outpatient`	Outpatient visits in the year before the encounter	Used
17	`number_emergency`	Emergency visits in the year before the encounter	Used

18	number_inpatient	Inpatient visits in the year before the encounter	Used
19	diag_1	Primary diagnosis (ICD-9 code)	Used
20	diag_2	Secondary diagnosis (ICD-9 code)	Used
21	diag_3	Additional secondary diagnosis (ICD-9 code)	Used
22	number_diagnoses	Number of diagnoses entered for the encounter	Used
23	max_glu_serum	Glucose serum test result: >200, >300, Norm, None (not measured)	Used
24	A1Cresult	HbA1c test result: >7, >8, Norm, None (not measured)	Used
25	metformin	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
26	repaglinide	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
27	nateglinide	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
28	chlorpropamide	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
29	glimepiride	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
30	acetohexamide	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
31	glipizide	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
32	glyburide	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
33	tolbutamide	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
34	pioglitazone	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
35	rosiglitazone	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
36	acarbose	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
37	miglitol	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
38	troglitazone	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
39	tolazamide	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
40	examide	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
41	citoglipton	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
42	insulin	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
43	glyburide-metformin	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
44	glipizide-metformin	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
45	glimepiride-pioglitazone	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored

46	`metformin-rosiglitazone`	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
47	`metformin-pioglitazone`	Whether this diabetes medication was prescribed and if the dose changed: No, Steady, Up, Down	Ignored
48	`change`	Whether any diabetes medication changed during the encounter: Ch, No	Ignored
49	`diabetesMed`	Whether any diabetes medication was prescribed: Yes/No	Used
50	`readmitted`	Readmission: NO, >30 (after 30 days), <30 (within 30 days)	Used

7. Possible identifiers and keys

- `encounter_id`: unique on all 101,766 rows. It is the primary key of the original table.
- `patient_nbr`: 71,518 distinct values. 16,773 patients have more than one encounter, so it identifies a separate **Patient** entity, not a row.
- `admission_type_id` (8 values), `discharge_disposition_id` (26), `admission_source_id` (17): codes into reference lists. Their meanings are published in the dataset's `IDs_mapping.csv`.
- `diag_1`–`diag_3`: 915 distinct ICD-9 codes, which identify a **Diagnosis** entity.

8. Missing, unknown and incomplete values

Attribute	Rows affected	Written as	Handling
`weight`	98,569 (96.9%)	`?`	Loaded as NULL
`max_glu_serum`	96,420 (94.7%)	`None`	NULL = test not measured
`A1Cresult`	84,748 (83.3%)	`None`	NULL = test not measured
`medical_specialty`	49,949 (49.1%)	`?`	Loaded as NULL
`payer_code`	40,256 (39.6%)	`?`	Loaded as NULL
`race`	2,273 (2.2%)	`?`	Loaded as NULL
`diag_3` / `diag_2` / `diag_1`	1,423 / 358 / 21	`?`	No diagnosis row is created for an empty slot
`gender`	3	`Unknown/Invalid`	Loaded as NULL

Inconsistencies found in the data:

- 123 patients have two different race values across their encounters, and 3 patients have two genders. Each patient keeps their most frequent non-missing value (ties go to the latest encounter).
- Age changes between encounters for 1,574 patients (birthdays between visits), so age is a property of the encounter, not the patient.
- In 5,690 encounters the same ICD-9 code appears in two diagnosis slots, so the diagnosis code alone cannot identify a diagnosis row; its slot position must.

Part 2. Normalization

Original relation

```
DIABETIC_DATA(encounter_id, patient_nbr, race, gender, age, weight, admission_type_id,
discharge_disposition_id, admission_source_id, time_in_hospital, payer_code,
medical_specialty, num_lab_procedures, num_procedures, num_medications, number_outpatient,
number_emergency, number_inpatient, diag_1, diag_2, diag_3, number_diagnoses, max_glu_serum,
A1Cresult, diabetesMed, readmitted)
```

- **Key:** `encounter_id`
- **Functional dependencies:** `encounter_id` → every other attribute; `patient_nbr` → race, gender; each admission/discharge/source code → its description.
- **Problems:** `diag_1`, `diag_2`, `diag_3` are a **repeating group**: three columns holding the same kind of value. Answering "which encounters had diagnosis 250.01?" means searching three columns, and a fourth diagnosis would need a new column. Patient data also repeats: race and gender are stored again on every encounter, 30,248 redundant copies (101,766 − 71,518).
- **Change needed for 1NF:** remove the repeating group.

First Normal Form (1NF)

Rule: every attribute holds a single atomic value, and there are no repeating groups.

```
ENCOUNTER_1NF(encounter_id, diag_position, icd9_code, patient_nbr, race, gender, age, weight,
admission_type_id, discharge_disposition_id, admission_source_id, time_in_hospital,
payer_code, medical_specialty, num_lab_procedures, num_procedures, num_medications,
number_outpatient, number_emergency, number_inpatient, number_diagnoses, max_glu_serum,
A1Cresult, diabetesMed, readmitted)
```

- **What changed:** the three diagnosis columns become one row per diagnosis, with `diag_position` = 1 (primary), 2 (secondary) or 3 (additional). An encounter with three diagnoses becomes three rows.
- **Key:** composite (`encounter_id`, `diag_position`). Position, not code, because 5,690 encounters repeat a code.
- **Functional dependencies:**
  - FD1: `encounter_id, diag_position` → `icd9_code` (depends on the *whole* key)
  - FD2: `encounter_id` → patient_nbr, age, weight, admission_type_id, discharge_disposition_id, admission_source_id, time_in_hospital, payer_code, medical_specialty, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed, readmitted (depends on *part* of the key)
  - FD3: `patient_nbr` → race, gender
  - FD4: `admission_type_id` → description; `discharge_disposition_id` → description; `admission_source_id` → description
- **Why it is 1NF:** every cell holds one value and the repeating group is gone.
- **Problem:** FD2 is a **partial dependency**. Twenty encounter attributes depend only on `encounter_id`, so each encounter's details are now copied up to three times, once per diagnosis row. Updating an encounter's length of stay would mean changing three rows, and missing one leaves the data inconsistent (an update anomaly).

- **Change needed for 2NF:** move every attribute that depends on only part of the key into its own table.

Second Normal Form (2NF)

Rule: the relation is in 1NF, and every non-key attribute depends on the *entire* primary key.

```
ENCOUNTER(encounter_id, patient_nbr, race, gender, age, weight, admission_type_id,
discharge_disposition_id, admission_source_id, time_in_hospital, payer_code,
medical_specialty, num_lab_procedures, num_procedures, num_medications, number_outpatient,
number_emergency, number_inpatient, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed,
readmitted)

ENCOUNTER_DIAGNOSIS(encounter_id, diag_position, icd9_code)   FK encounter_id → ENCOUNTER
```

- **What changed:** FD2's attributes moved to `ENCOUNTER`, keyed by `encounter_id` alone. `ENCOUNTER_DIAGNOSIS` keeps only what needs the full composite key.
- **Why it is 2NF:** `ENCOUNTER` has a single-column key, so no attribute can depend on part of it. In `ENCOUNTER_DIAGNOSIS`, the only non-key attribute, `icd9_code`, needs both `encounter_id` and `diag_position`. No partial dependencies remain.
- **Remaining redundancy:** FD3 and FD4 are *transitive* dependencies (`encounter_id` → `patient_nbr` → race). Race and gender still repeat for every encounter of the same patient, and the admission codes have no descriptions stored anywhere.

Final design: removing the remaining redundancy

The assignment requires 2NF and asks the design to minimize redundancy, so we split out the two remaining dependencies. This keeps every table in 2NF (and also satisfies 3NF):

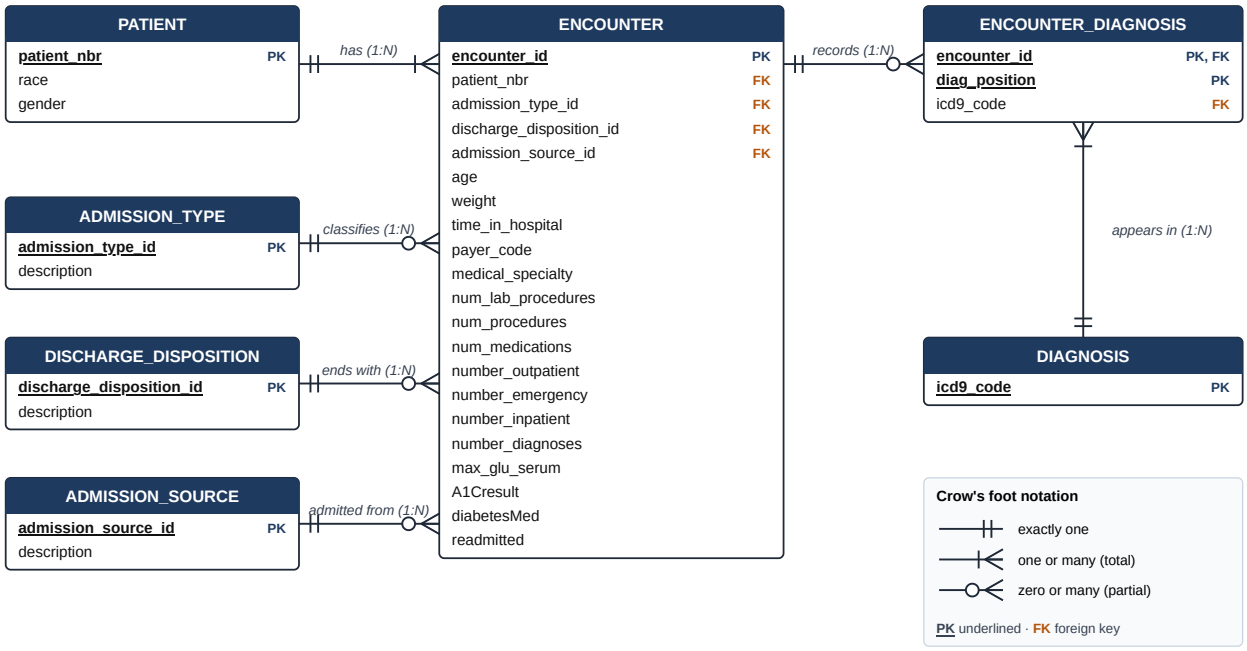
- FD3 → `PATIENT`(patient_nbr, race, gender). Race and gender are stored once per patient instead of once per encounter.
- FD4 → three reference tables (`ADMISSION_TYPE`, `DISCHARGE_DISPOSITION`, `ADMISSION_SOURCE`) whose descriptions come from `IDs_mapping.csv`.
- `DIAGNOSIS`(icd9_code) lists each code once, so `ENCOUNTER_DIAGNOSIS.icd9_code` can be a foreign key.

Table	Attributes	Primary key	Foreign keys	Rows
`patient`	patient_nbr, race, gender	patient_nbr	none	71,518
`admission_type`	admission_type_id, description	admission_type_id	none	8
`discharge_disposition`	discharge_disposition_id, description	discharge_disposition_id	none	26
`admission_source`	admission_source_id, description	admission_source_id	none	17
`diagnosis`	icd9_code	icd9_code	none	915

`encounter`	encounter_id, patient_nbr, admission_type_id, discharge_disposition_id, admission_source_id, age, weight, time_in_hospital, payer_code, medical_specialty, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed, readmitted	encounter_id	patient_nbr → patient admission_type_id → admission_type discharge_disposition_id → discharge_disposition admission_source_id → admission_source	101,766
`encounter_diagnosis`	encounter_id, diag_position, icd9_code	(encounter_id, diag_position)	encounter_id → encounter icd9_code → diagnosis	303,496

Nothing from the 26 attributes is lost: every value in the original rows can be rebuilt by joining these tables. The row counts are the targets for verification in Part 7 (303,496 = 3 × 101,766 diagnosis slots minus 1,802 empty ones).

Part 3. Entity-Relationship Diagram



Relationship	Type	Participation
Patient *has* Encounter	1:N	Total on both sides: every encounter belongs to exactly one patient, and every patient in the data has at least one encounter.
Admission Type *classifies* Encounter	1:N	Every encounter has exactly one admission type (total). A type may have no encounters (partial).
Discharge Disposition *ends* Encounter	1:N	Same as above.
Admission Source *admits* Encounter	1:N	Same as above.
Encounter *has* Diagnosis	**M:N**	An encounter has 0–3 diagnoses (partial: 21 encounters have no primary diagnosis). A diagnosis code appears in one or more encounters (total). Resolved by the associative entity ENCOUNTER_DIAGNOSIS, which also records each diagnosis's position.

There is no 1:1 relationship in this design. The diagram matches Part 2 exactly: seven entities for seven tables, and each relationship line is one foreign key.

Part 4. Relational Schema

Primary keys are **underlined**; foreign keys are in *italics*.

```
patient(patient_nbr, race, gender)
admission_type(admission_type_id, description)
discharge_disposition(discharge_disposition_id, description)
admission_source(admission_source_id, description)
diagnosis(icd9_code)
encounter(encounter_id, patient_nbr, admission_type_id, discharge_disposition_id,
admission_source_id, age, weight, time_in_hospital, payer_code, medical_specialty,
num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency,
number_inpatient, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed, readmitted)
encounter_diagnosis(encounter_id, diag_position, icd9_code)
```

How the ER relationships are represented

ER relationship	Type	Representation in the schema
Patient has Encounter	1:N	`encounter.patient_nbr` → `patient.patient_nbr`. The foreign key goes on the "many" side and is NOT NULL because participation is total.
Admission Type classifies Encounter	1:N	`encounter.admission_type_id` → `admission_type`, NOT NULL
Discharge Disposition ends Encounter	1:N	`encounter.discharge_disposition_id` → `discharge_disposition`, NOT NULL
Admission Source admits Encounter	1:N	`encounter.admission_source_id` → `admission_source`, NOT NULL
Encounter has Diagnosis	M:N	A separate table, `encounter_diagnosis`, with a foreign key to each side (`encounter_id` → encounter, `icd9_code` → diagnosis). Its primary key (`encounter_id`, `diag_position`) allows up to three diagnoses per encounter and keeps their order; `diag_position` is limited to 1–3.

Creation order (needed so every foreign key has a parent): patient, admission_type, discharge_disposition, admission_source, diagnosis → encounter → encounter_diagnosis.